

AAROHAN 2026

Ascent of Knowledge

X

Sample
Paper

Name

Roll No.

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BITS Pilani
Pilani Campus

APOGEE2026

Instructions: -

- Read the passage carefully, it contains all the necessary details for solving the questions that follow it.
- The questions will be based on Physics, Chemistry, Mathematics and Analytical Skills.
- The questions can test your understanding of one or more of the above subjects at once.
- Questions marked with an asterisk(*) **may** have more than one option as the correct answer(s). You must select all correct option(s) to earn marks.
- **All other questions have only one correct alternative.** The marking scheme is
 - +3 -> Correct Answer
 - 0-> Unattempted Question
 - -1 -> Incorrect Answer
- For **multiple correct answer** type questions the marking is scheme is
 - +4 -> All correct options selected
 - 0 -> All other cases
- If an assumption is to be made, it will be mentioned in the question itself, along with all other necessary data/information required. Try to crack with the information that has been given- we test your **thinking process** not rote knowledge.
- Before you start this exam, **calm down** and remind yourself that **you can crack this! Keep moving forward, do not get stuck on one question.**
- **Best of Luck!!**

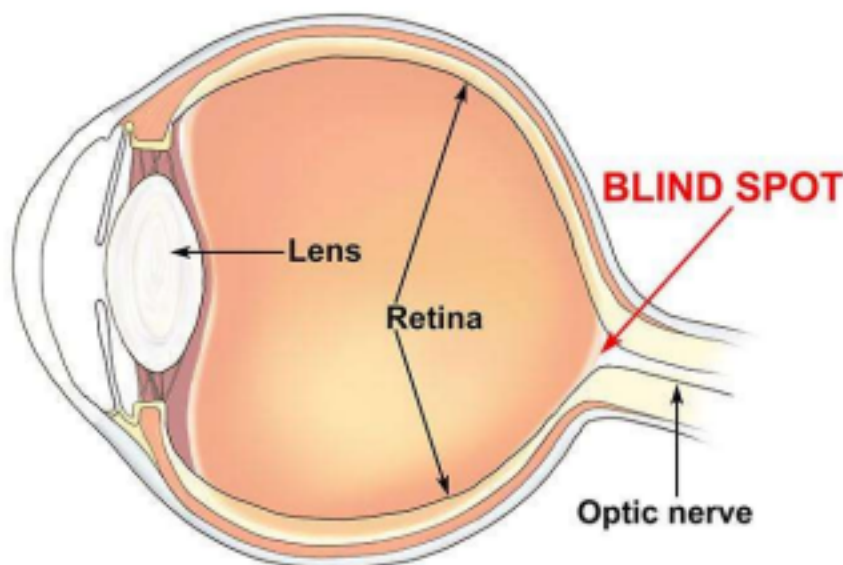


INVISIBILITY

Invisibility is the state of an object that cannot be seen. An object in this state is said to be invisible. There are various methods by which we can render invisibility to an object. Most of these methods are mere optical illusions because ultimately the state of being seen or not depends upon how cleverly the view of the observer can be evaded.

BLIND SPOT:

Blind spot is a small portion of the visual field of each eye that corresponds to the position of the optic disk (also known as the optic nerve head) within the retina. There are no photoreceptors (i.e., rods or cones) in the optic disk, and, therefore, there is no image detection in this area. The blind spot of the right eye is located to the right of the centre of vision and vice versa in the left eye. With both eyes open, the blind spots are not perceived because the visual fields of the two eyes overlap. Indeed, even with one eye closed, the blind spot can be difficult to detect subjectively because of the ability of the brain to “fill in” or ignore the missing portion of the image.



CAMOUFLAGING:

Camouflage is the use of any combination of materials, coloration, or illumination for concealment, either by making animals or objects hard to see (crypsis), or by disguising them as something else (mimesis). Examples include the leopard's spotted coat, the battledress of a modern soldier. The majority of camouflage methods aim for crypsis, often through a general resemblance to



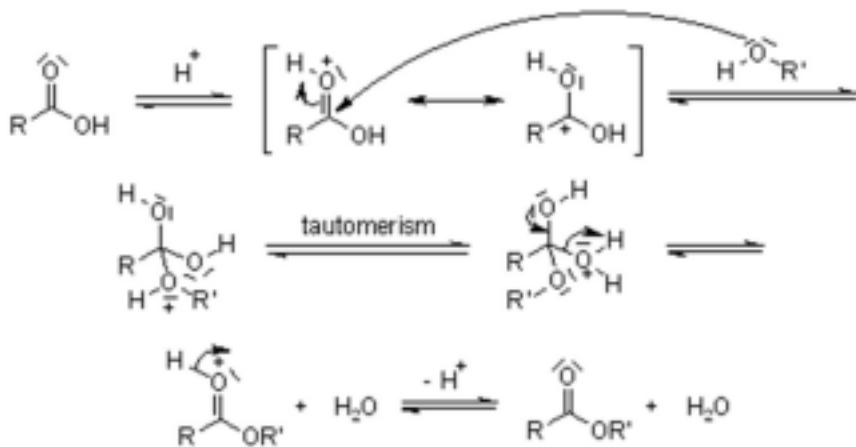
the background, high contrast disruptive coloration, eliminating shadow, and countershading. Some animals, such as chameleons and octopuses, are capable of actively changing their skin pattern and colours, whether for camouflage or for signalling. It is possible that some plants use camouflage to evade being eaten by herbivores.

BEING INVISIBLE BY BEING COLORLESS!

Since objects can be seen by light in the visible spectrum from a source reflecting off their surfaces and hitting the viewer's eye, the most natural form of invisibility (whether real or fictional) is an object that neither reflects nor absorbs light (that is, it allows light to pass through it) or if it absorbs all the light that falls on it. Perfect light absorption occurs in the heated metal because light rays emerging from the oxide layer and the metal substrate come together in such a way that they cancel each other out.

ESTERIFICATION AND BASIC MECHANISM

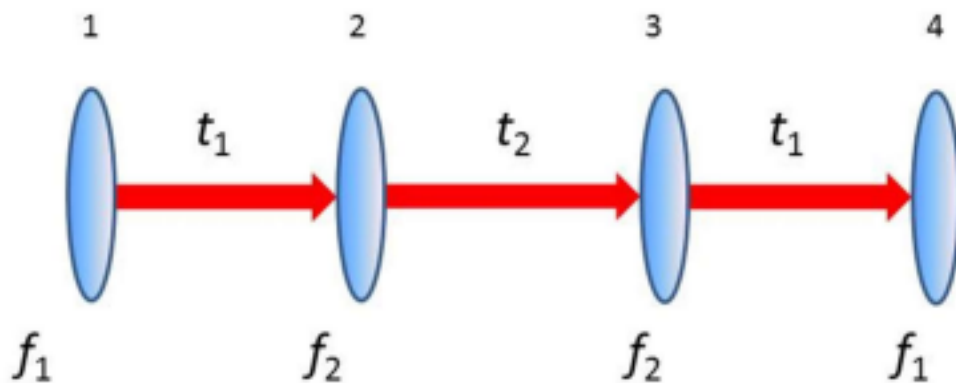
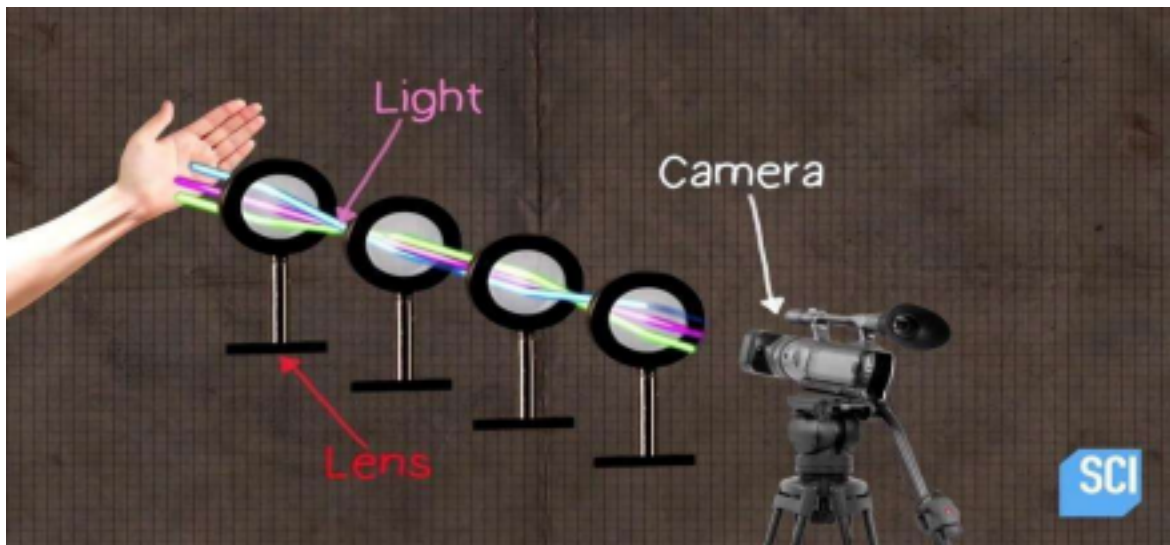
Esterification is a chemical reaction that occurs between the acid (usually carboxylic acid) and the alcohol (or compounds containing the hydroxyl group) where esters are obtained. The reaction takes place in acidic environments. In this process, water is also obtained. It, therefore, falls into the category of “condensation reactions”. Most carboxylic acids are suitable for the reaction, but the alcohol should generally be a primary or secondary alkyl. The basic reaction mechanism of the esterification reaction is as described below-



ROCHESTER CLOAK

Rochester Cloak is an optical device which uses 4 lenses to control light in such a way that it renders an object invisible. When the light enters the first lens, it flips the image upside down. The second lens sends the light straight into the 3rd lens. Since the light is focused into a point, any object placed around that point cannot be seen when looking through the lenses. The light is essentially bending around

the object. Once the light hits the point just past the 3rd lens, it then flips back right side up. This allows the background image to pass through all of the lenses without warping or being seen as upside down. The object to be cloaked is kept in between lenses 2 and 3 in the below diagram.



QUESTIONS:-

Q.1 For a Rochester Cloak described in the passage, if value of f_1 is 40cm , f_2 is 60 cm and the separation between the lenses(t_1) is kept as 100cm then for a circular background light emitting source of radius 5cm and the aperture of the lenses as 20cm, find the area of the cloaking region? (the region between the lenses 2 and 3 in which an object is kept would be invisible)

- a. 50π
- b. 150π
- c. 125π
- d. 31.25π

Q.2 For the same Rochester cloak arrangement as described in Q.1 what is the maximum length of an object which can be cloaked(made invisible) completely by the arrangement?

- a. 5 cm
- b. $10\sqrt{5}$ cm
- c. $5\sqrt{10}$
- d. 10 cm

Q3. Which of the following could be the reasons that in spite of having a blind spot in our eyes we are able to see the complete view in front of our eyes?

- a. The brain has the ability to “guess” the image which could be present at the blind spot region and therefore no blank spot in vision is felt.
- b. The reflection of the light from our eye lens takes place in such a way that the image is never formed on the blind spot region.
- c. The blind spots are such that the image formed at the blind spot of the right eye is captured by the left eye and hence no “blank spot” in vision is observed.
- d. None Of these.

Q.4 Which of the following phenomenon(s) could render invisibility to an object kept in a surrounding medium of refractive index μ ?

- i. By making the refractive index of the object as μ .
 - ii. By making the object a perfect absorber of light.
 - iii. Camouflaging the object.
 - iv. If the image of the object is formed on the blind spot of either of the eyes.
- a. (i), (ii) and (iv)
 - b. (i) only
 - c. (i) and (iii).
 - d. (i), (ii), (iii) and (iv).

Q.5 In conquest to attain invisibility Mr.X and Mr.Y both are in search of two potions A and B. But soon Mr.X and Mr.Y come to know that the potions don't react properly every time and are such that A works 3 out of every 4 times it is used and B works only 1 out of every 6 times used. Also, Mr.X has twice as much as chance as Mr.Y in finding A. Mr.Y has thrice as much as chance as Mr.X in finding B. Who out of Mr.X and Y is more likely to make a potion that works? (Only one of Mr. X and Mr. Y can procure potion A and B)

- A. Mr. X B. Mr. Y C. Both have equal chances D. Information is insufficient

Q.6* The synthesis of the invisibility potion requires an esterification reaction between the two potions A and B where B is alcohol consisting of O18 of oxygen as its oxygen atom. Which of the following statement(s) is/are true for the reaction?

- a. The reaction is an example of a dehydration reaction
- b. A is acting as a base and B is acting as an acid in the reaction.
- c. The water molecule released in the reaction consists of the oxygen atom of A.
- d. Both the oxygen atoms in the product molecule are O16 isotopes.

Q.7* In search of 2 important ingredients A and B for the invisibility potion you meet a mad scientist who has both of the ingredients but puts up a riddle that if the next statement made by you is true then he would give you A and if it is false then he would give you B. With which of the following statements would you get both the ingredients?



- a. You would give me A.
- b. You would not give me A.
- c. You would give me B.
- d. You would not give me B.

Q8. "Sometimes the things that seem invisible are right in front of us" A superhuman enters a bank in pursuit of a diamond tiara kept inside a vault. He sneaks past the security being invisible but is stuck at the door. He needs a password to go ahead. He discovers that the machine reads passwords in a coded language where OWB stands for PEN, UBJ stands for INK and GYBR stands for HUNT. Suppose the password of the machine is BRUISE, what should he enter to unlock the vault?

- a. VEYUAW
- b. YIFRHV
- c. VHRFIY
- d. WAUYEV



DEFENSE

“Forts are either fortified by nature, by hand, or by both, which is considered stronger.”

Castles play an instrumental role during wars.....

To build a castle, we need to study all the factors that need to be considered. The most important factor while building a fort is location.

A fort should always be built on a dominating landscape, with a good view in all directions. Enemies should find it hard to reach the fort and thus rivers and slopes can be used as an advantage. At the same time, it should also be taken care that materials are available nearby and can be easily

transported to the

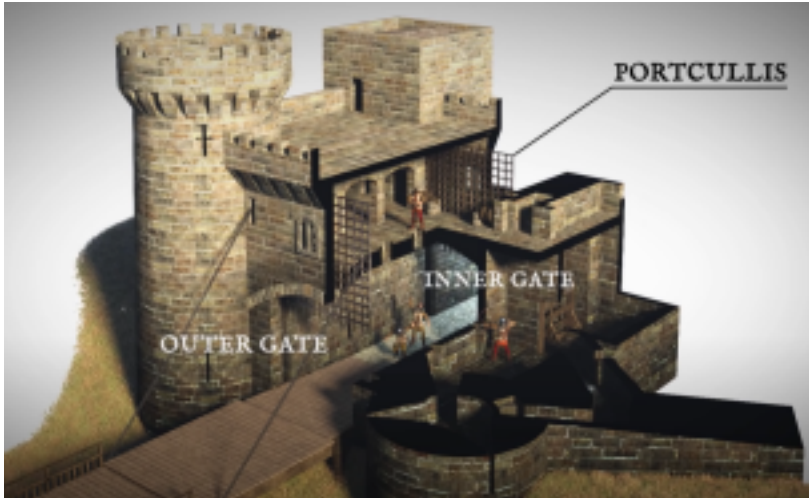
site, as transportation in the medieval period was a huge challenge and expensive.

Once we finalize a location, we start to build our fort.

The first component of the fort is the keep, the strongest portion of fortification of a castle. It can be rectangular, square, or circular. It is usually 2-4 stories high having 6 feet thick walls. It also has towers at each corner and the forebuilding (the entrance) is at the 1st floor. In a keep are private apartments, chapel, halls for dining and other entertainment purposes, cellar, etc. It also acts as a place for last refuge.

Then we build a curtain wall, to enclose the keep and other important buildings like stables, stores, hall, forge, etc. This enclosed area is called Bailey. The curtain wall should have crenellations, as shown below, which help protect the garrison soldiers during an attack. The curtain wall should always have Flanking Towers at regular intervals. These towers may be Square or Circular or D-shaped. These towers should always be projected out of the curtain wall. The curtain wall should also have Loopholes which are arrow slits through which soldiers can shoot their arrows.

Now coming to the entrance of the fort. The entrance of the fort should be protected using a gatehouse. It contains flanking towers on both the outer corners. It should contain two gates, inner and outer, designed to take some amount of force. The gatehouse should also contain preferably two or more Portcullis (at least one) to trap the enemies in a kill zone. Then the soldiers present in the gatehouse can kill the trapped enemies using murder holes in walls and ceiling (usually 2 archers). A drawbridge is also necessary which can be pulled up to stop the enemy from entering the castle.

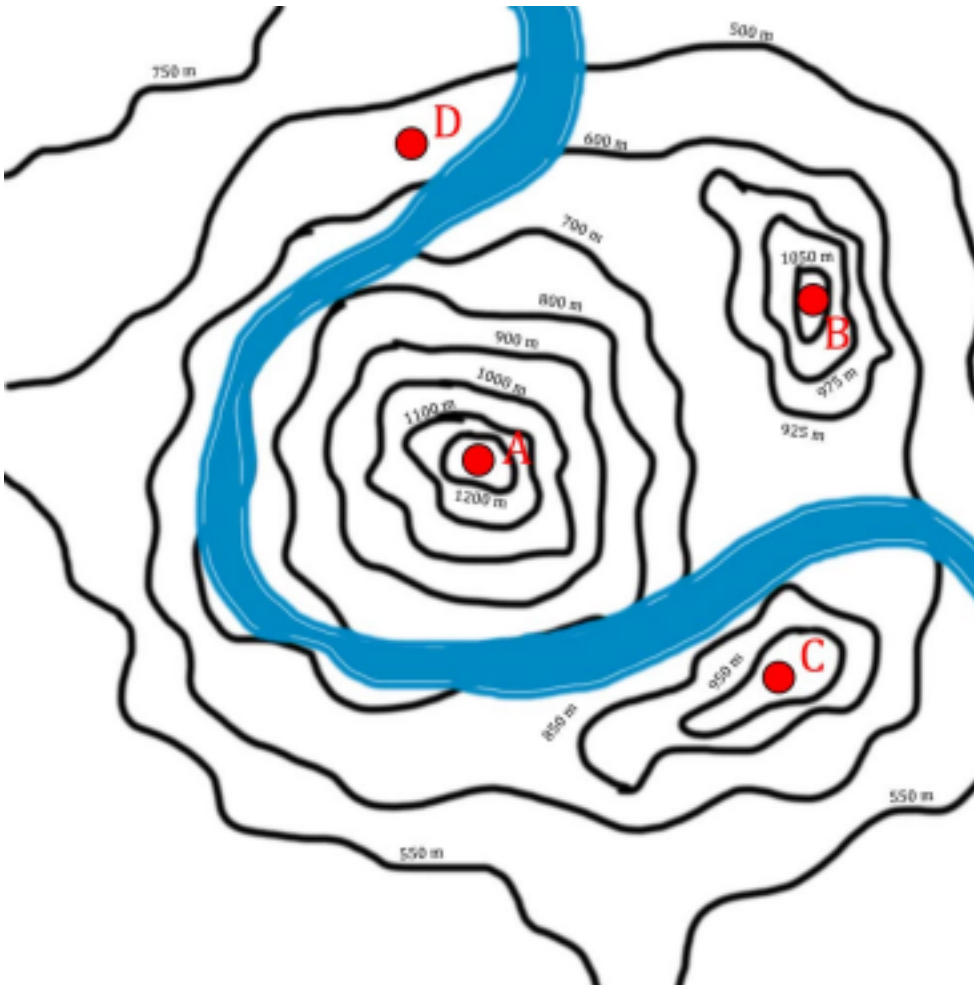


To make our castle more imposing, an outer curtain wall should also be present.

Contouring is the determination of elevations of various points on land and fixing these on a same horizontal position in the contour map. Contouring is done with the help of a contour line. First we find elevations of various points in an area and take a note of them on the map. Then we use a contour line to join those. A contour line is a line that passes through the points having the same elevation. If contour lines are evenly spaced, then they indicate an area having uniform slope. Closed contours are either a hill or a depression depending on how the elevation is changing. The distance between these contour lines indicates the steepness of the slope, the greater the distance between two lines the less the slope and vice-versa.

Questions-:

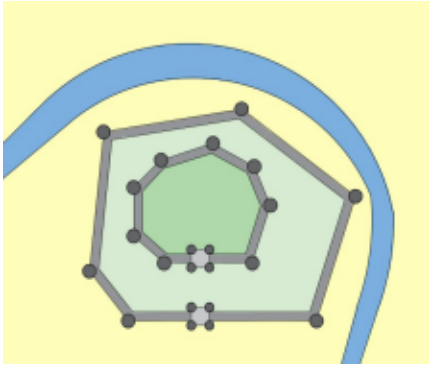
1. With your knowledge of contouring, which of these locations would be best for building our castle?



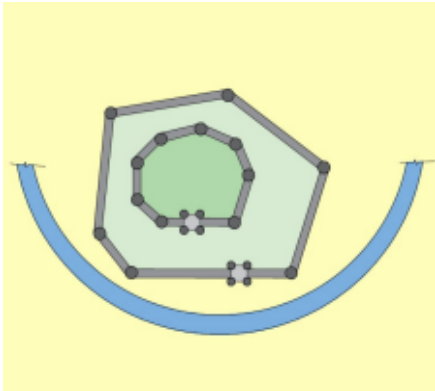
- a. A
- b. B
- c. C
- d. D

2. Which of the following Castle designs is the most helpful design wrt defense?

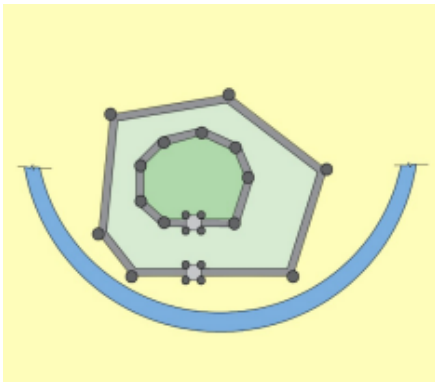
a.



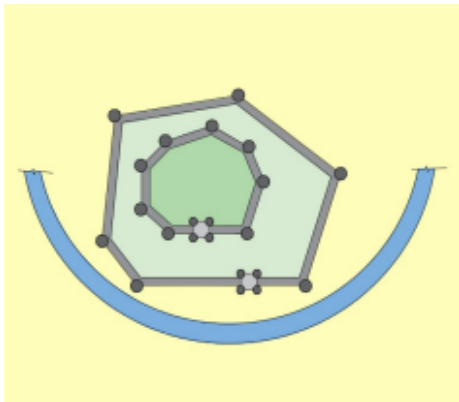
b.



c.

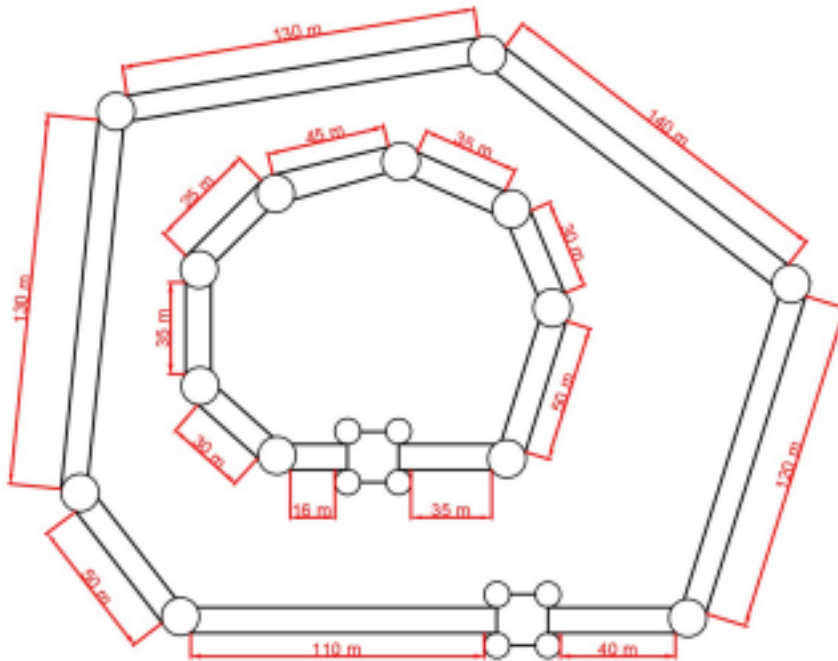


d.



3. You need to train archers in order to defend your fort successfully (dimensions of both types of castles are given below). You wish to have archers in a position on the 9 m high curtain walls such that they can observe in all directions. The field of view of archers is 60° and they are equally strong on both sides. The 2 archers on each watchtower cover the area that is not being covered by those standing on the curtain walls, while there are archers stationed in the gatehouse also along with an archer each on the gatehouse's watchtower. Find the minimum number of archers required to satisfy our condition considering that there is one archer at the center of every wall between watchtowers. Assume that the bow imparts a momentum of $1.35 \text{ kg}\cdot\text{m/s}$ to a 60 g arrow and $\sqrt{5} = 2.24$.

P.S.: The orientation of the watchtower does not affect the dimension between two watchtowers (You may use help of the information given in previous passages)



a. 75

b. 73

c. 69

d. 65

4. Which of the following shapes of the loopholes would be best suited for an archer inside the castle.

a.



b.



c.

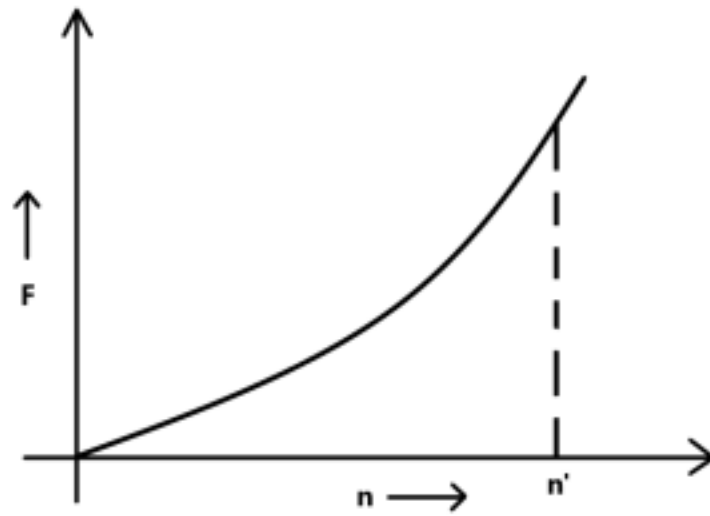


5. Why are round-watchtowers preferred over square watchtowers?

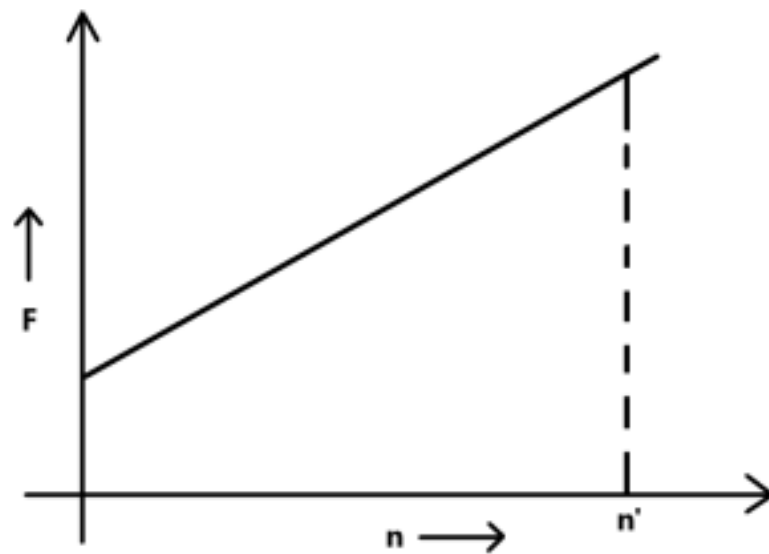
- a. Round watchtowers can store more ammunition in the same space as compared to square ones
- b. Square watchtowers cover a lesser field of view.
- c. Square watchtowers are more vulnerable due to corners.
- d. Round watchtowers provide a much more aesthetically pleasing look.

6. Which of the following graph gives the relation between the force that the chain is applying on the drawbridge to pull it up and the number of rounds (n) the pulley has completed. Assume that the change that the chain remains completely taught in the process. You can assume that the length of the drawbridge is around 300 times the radius of the pulley, n' is the number of full rounds of the pulley.

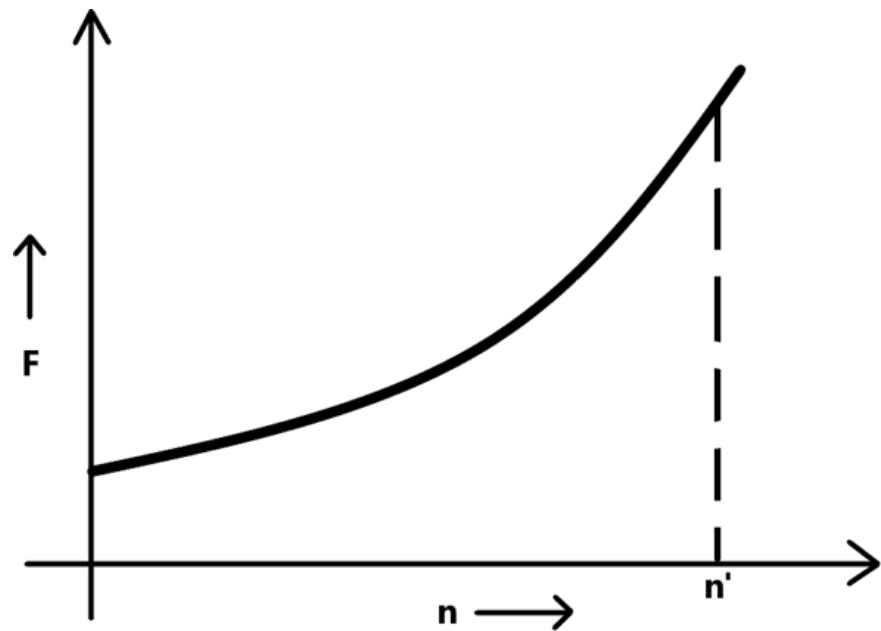
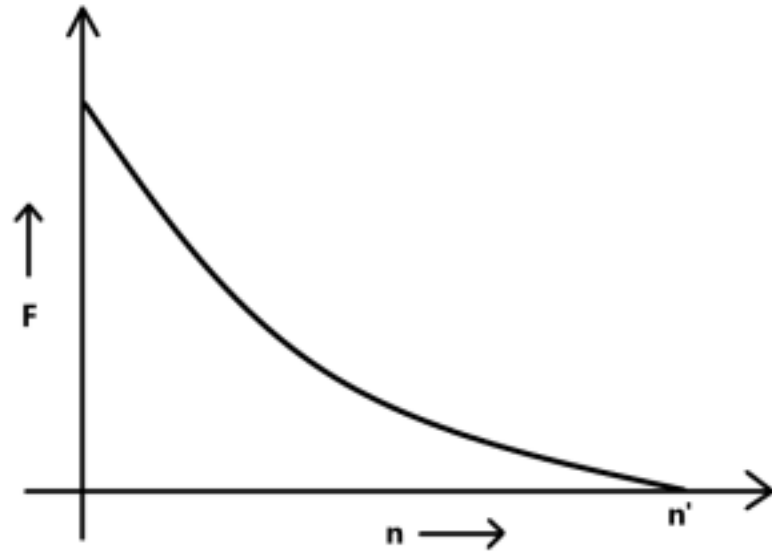
a.



b.



c.



d.

7. You find information about a survey of the people staying in the castle. In the castle 40 common people, different people use their savings for different facilities. 19 people use their savings for farming, 24 use it for animal husbandry and 16 use it for investing. Moreover, it was also found that 6 people use the boat only for investing, 4 of them use it only as a means of investing and farming, 7 of them use it only for farming and animal husbandry. Find out the number of people who use their savings for all three purposes.



- a. 2
- b. 3
- c. 4
- d. 5

8. While looking for the remains of war in a castle you find a treasure chest which has a 1 digit pass code. It is written on a paper "Last digit of 1989^{1989} ". What is the pass code?

- a. 1
- b. 3
- c. 7
- d. 9